1)

$$e = Bl(u_{coil} - u_{ground})$$

$$m_{coil}\, a_{coil} = F_{coil,net}$$

$$a_{coil} = j\omega u_{coil}$$

$$x_0 = \frac{u_0}{j\omega}$$

$$F_{coil,net} = k_{eq}(x_{ground} - x_{coil})$$

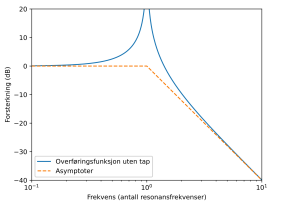
$$= \frac{k_{eq}}{j\omega}(u_{ground} - u_{coil})$$

a) $m_{coil}\, a_{coil} = F_{coil,net}$

$$m_{coil}\, j\omega u_{coil} = \frac{k_{eq}}{j\omega}(u_{ground} - u_{coil})$$

$$\frac{u_{coil}}{u_{ground}} = \frac{\frac{k_{eq}}{j\omega}}{\frac{k_{eq}}{j\omega} + j\omega m_{coil}}$$

$$= \frac{1}{1 - \frac{\omega^2}{\omega_0^2}}, \quad \omega_0 = \sqrt{\frac{k_{eq}}{m_{coil}}} = \text{resonansfrekvens}$$




b) $\left|\frac{u_{coil}}{u_{ground}}\right| = 1$

c) $$e = Bl \cdot u_{ground}\left(\frac{1}{1 - \frac{\omega^2}{\omega_0^2}} - 1\right)$$

$$= Bl \cdot u_{ground} \frac{1}{\frac{\omega_0^2}{\omega^2} - 1}$$

2)

$$R(T) = R(0)[1 + AT + BT^2]$$

$$R_{ground}(T) = R(0)[1 + DT]$$

a) $R(T) = R(0)[1 + 0.1T + 0.2T^2]$

$= R(0)[1 + DT]$, $D = 0.1T$ → $T = 100$

b) $E(T) = R_0(D - A) + \left(1 - \frac{T}{100}\right)$

$E(0) = R_0(D - A)\,25$

$E(200) = R_0(D - A)\,200 = 11.05\,\Omega$

c) $C + DT_1 = R(0)(1 + AT_1 + T_1^2 B)$

$C + DT_2 = R(0)(1 + T_2 + T_2^2 B)$

$$E(T) = C + DT - R(0)[1 + AT + BT^2]$$

$$= -BR(0)[T_1T_2 - (T_1+T_2)T + T^2]$$

$$= -BR(0)(T - T_1)(T - T_2)$$

$E(500) \approx 6.2\,\Omega$

$$T = \frac{-A \pm \sqrt{A^2 - 4B(C - \frac{E}{R(0)})}}{2B}$$